

HARMONIZE

Earth **O**bservation **D**ata **C**ubes tuned for **H**ealth **R**esponse **S**ystems (EODCtHRS)

Cube4Health

Marcos L. Rodrigues on behalf of the
HARMONIZE-INPE Team



Date: 2026/05/04



Ministério da Saúde

FIOCRUZ

Fundação Oswaldo Cruz



**Barcelona
Supercomputing
Center**

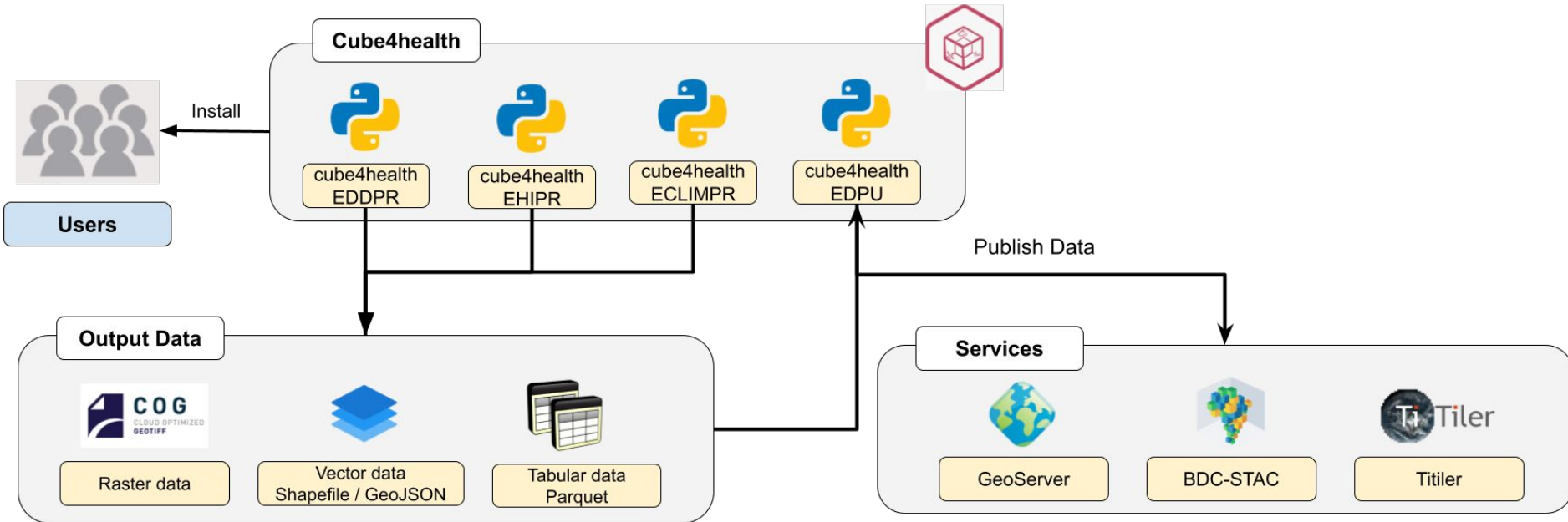
Centro Nacional de Supercomputación



This work was supported by the Wellcome Trust grant number 224694/Z/21/Z.

cube4health

Objective: integrate a set of algorithms from individual packages as modules in a **general package** (cube4health) that will process raw data from Drones, Climate, and Health to produce Cloud Optimized Images, Spectral Indices (NDVI), and Indicators.



HARMONIZE INPE Team



Miguel Monteiro
*INPE Project Coordinator
Data Cube*



Lubia Vinas
Team Coordinator



Karine Ferreira
Team Coordinator



Gilberto Queiroz
Team Coordinator



Isabel Escada
Drone Team Coordinator



Sidnei Sant'Anna
Drone Team Member



Claudia Santos
Drone Team Member



Adeline Maciel
*IT Team Member
Climate module*



Marcos Rodrigues
*IT Team Member
Drone module*



Yuri Domaradzki
*IT Team Member
Health module*



Luana Becker
*IT Team Member
BDC Lab*



Gabriel Sansigolo
*IT Team Member
Harmonize explorer
and Harmonize DS*



João Pedro Diehl
BDC Lab



Ana Rorato
Drone Team Member



Ana Dal'Asta
Applications team



“O objetivo principal é desenvolver ferramentas digitais econômicas e reproduzíveis para as partes interessadas em hotspots de mudanças climáticas na América Latina e no Caribe, incluindo cidades, pequenas ilhas, terras altas e a floresta Amazônica.”



Dominican
Republic

Cajamarca,
Tolima
(Colombia)

Cametá, Mocajuba
Baixo Tocantins
(Brazil)

Iquitos, Loreto
(Peru)

Patos, Mãe d'Água
Semiárido - Paraíba
(Brazil)

Brazil Hotspots

Wider community of practice

City toolkit

Application:
seasonal dengue early
warning system

Stakeholder:
Infodengue, Brazil



**Semiarid
(Patos-PB)**

Satellite images,
gridded climate
reanalysis and
forecasts

Drone images
and weather
station data



HARMONIZE

Wider community of practice

Amazon rainforest toolkit

Application:
landscape change
and vector-borne
disease emergence

Stakeholder:
Regional Health
Directorate of Loreto, Peru



**Lower Tocantins
(Mocajuba and Cametá -PA)**



Drone imagery



Health data

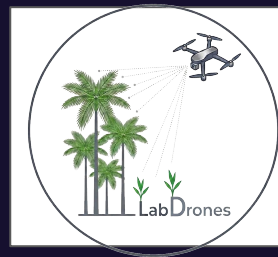


Climate data

Protocol for Drone Use in Health: From Mapping to Data Integration in Cube4Health

Ana Rorato
Maria Isabel S. Escada
Sidnei Sant'Anna
Marcos Rodrigues
Claudia Santos

INPE - Brasil



Brazil hotspots

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seasonal dengue early
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HARMONIZE

Wider community of practice

Amazon rainforest toolkit

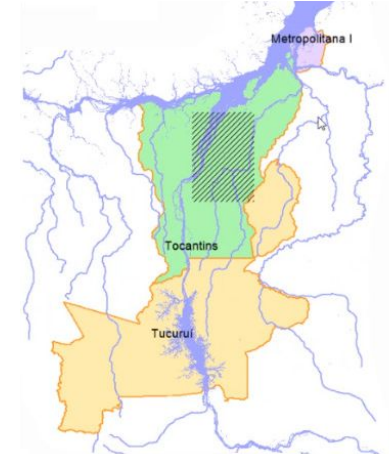
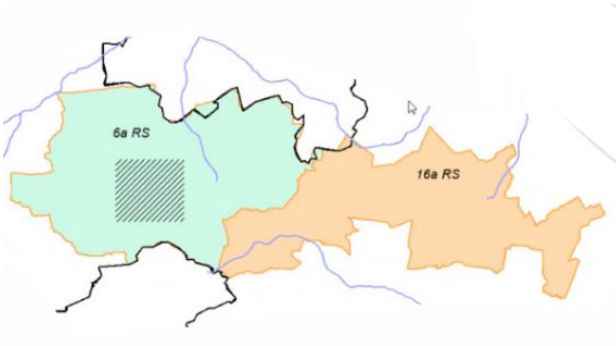
Application:
landscape change
and vector-borne
disease emergence

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Regional Health
Directorate of Loreto, Peru



Semiarid
(Patos-PB)

Amazon
(Baixo Tocantins - PA)



What is the objective of the mapping?



To map landscapes associated with vector-borne diseases



What are the target objects?

Elements in the landscape or environmental disturbances that are associated with the **proliferation of disease vectors.**



Dengue



Malaria



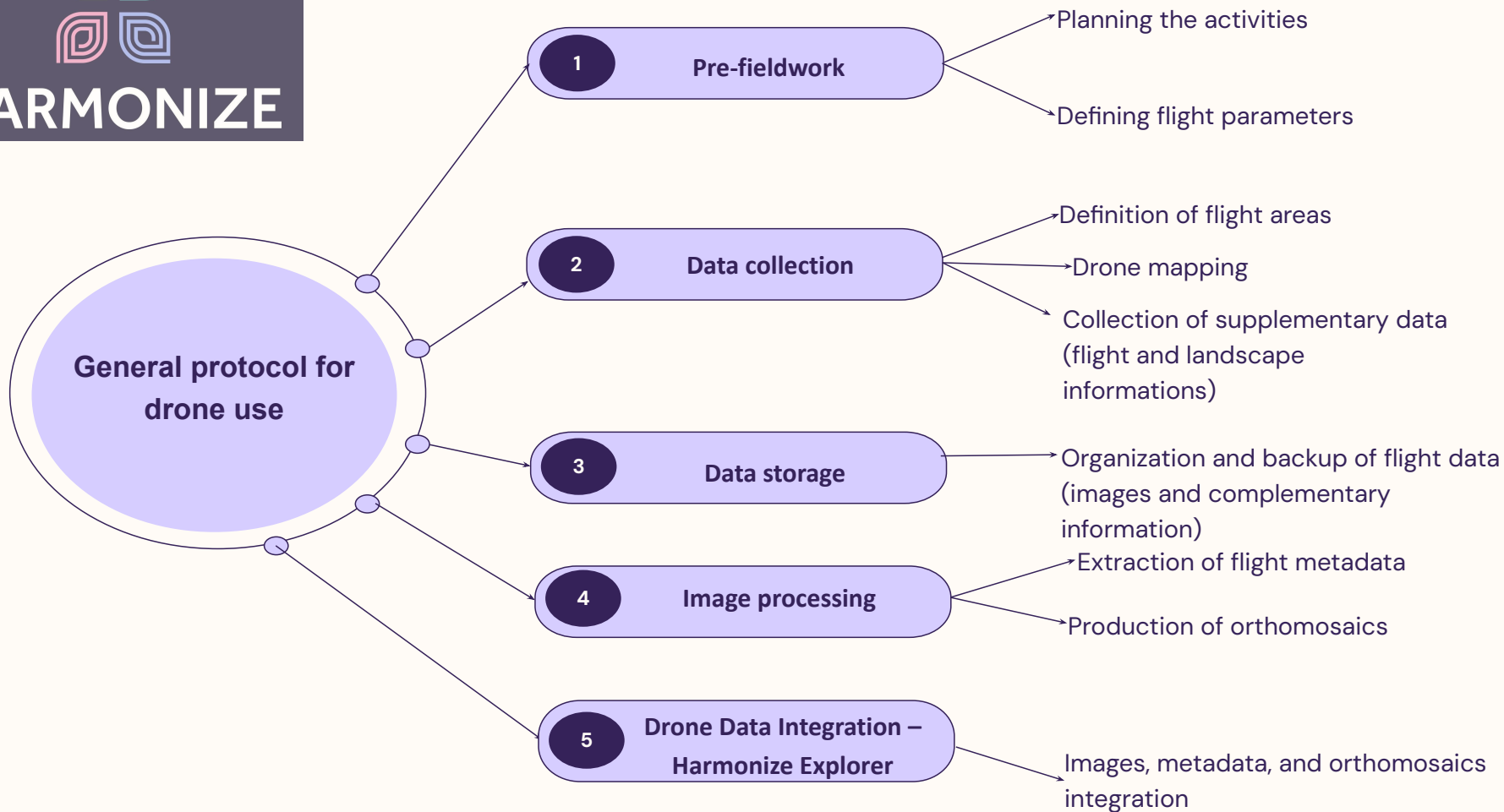
Chagas



Leishmaniosis



HARMONIZE





Camera

RGB

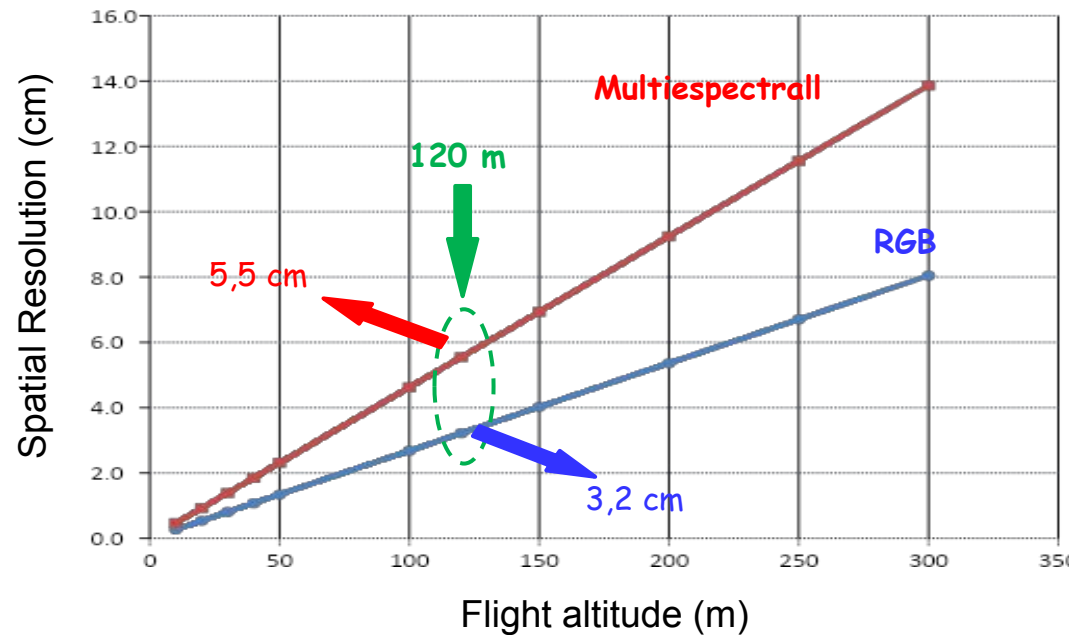
Multispectral

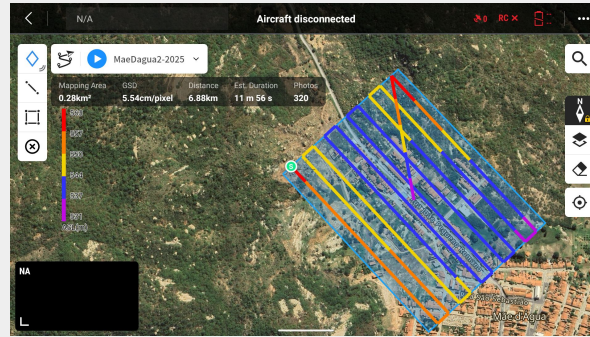
(Green (G): 560 ± 16 nm;

Red (R): 650 ± 16 nm;

Red Edge (RE): 730 ± 16 nm;

Near-Infrared (NIR): 860 ± 26 nm.



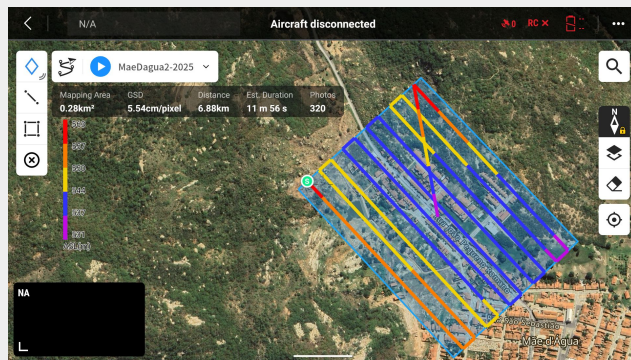


Orthomosaic Generation from Drone Flight Images:

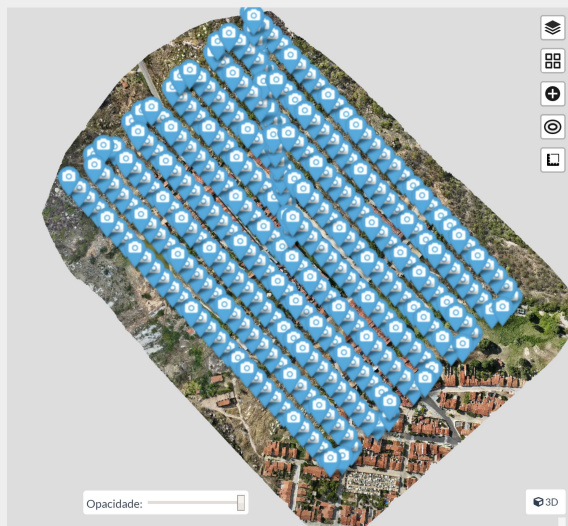


OpenDroneMap (ODM) is an open-source platform for processing aerial images captured by drones, generating geospatial products such as orthomosaics, digital terrain models, and 3D point clouds.

Flight



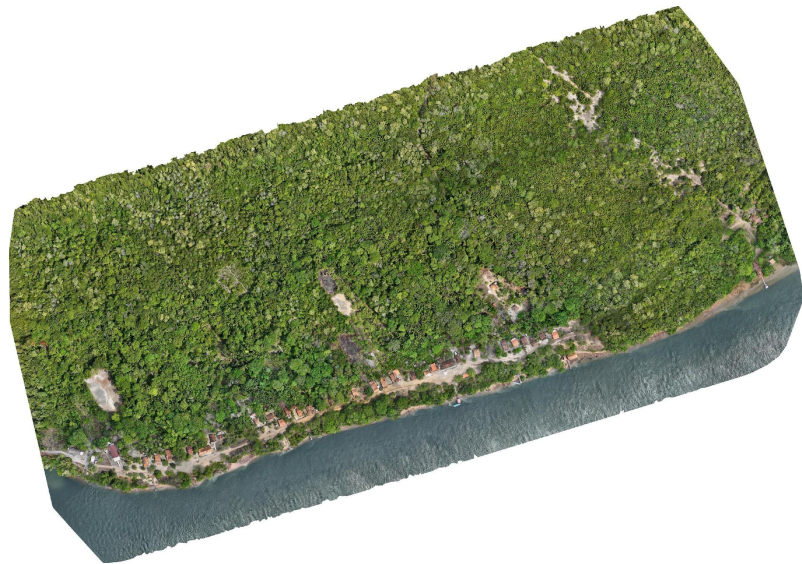
It uses the overlap between the images to find homologous points.



Orthomosaic



Mangabeira riverine community



Flight duration: 22 min 38 s

Mapped area: 54,21 ha

Images: 3,074

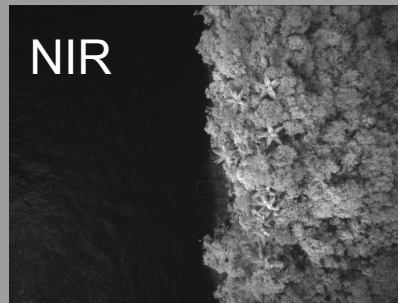
Raw data volume: ~29 GB

RGB

614 RGB JPEG files



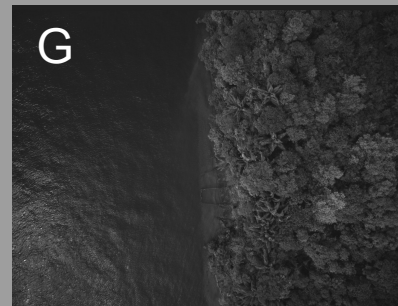
NIR



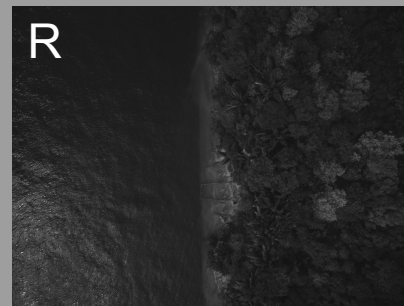
RE



G



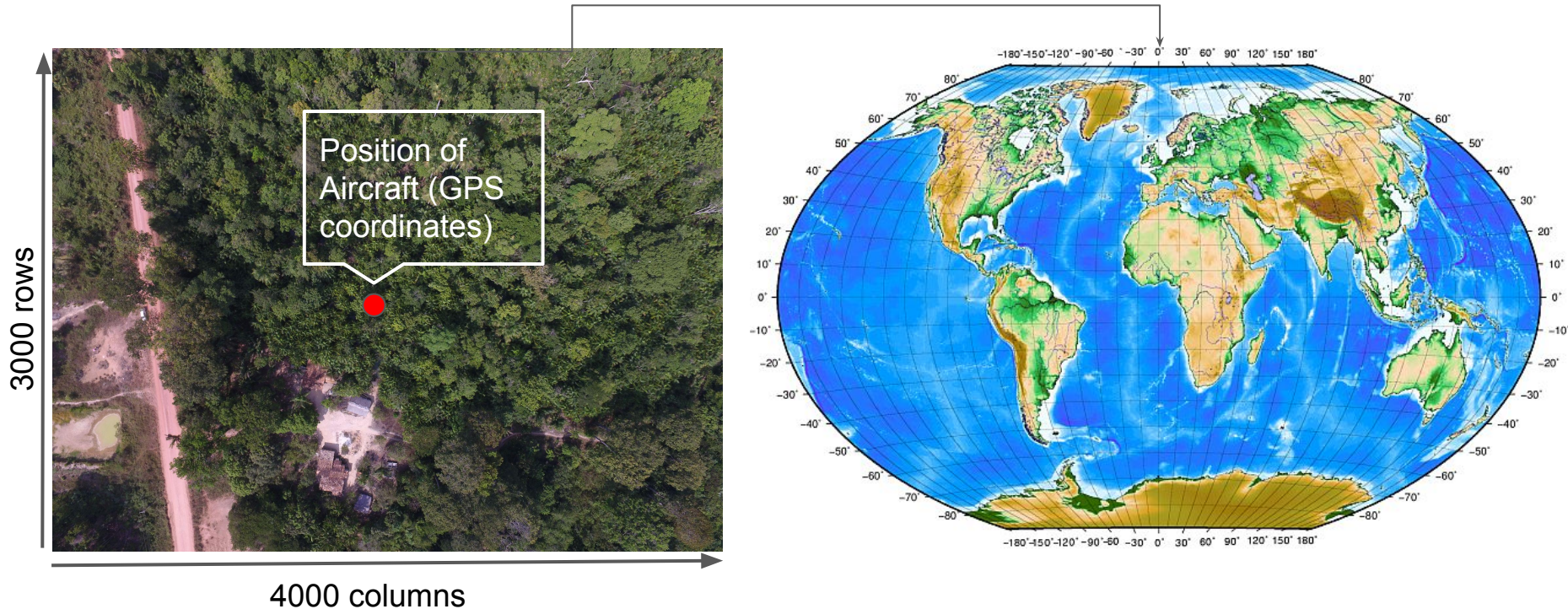
R



2,456 multispectral TIFF files

Drones images georeferencing for integration with STAC

“A spatiotemporal asset is any file that represents information about the Earth at a certain place and time.”



Drone images - Integration with STAC

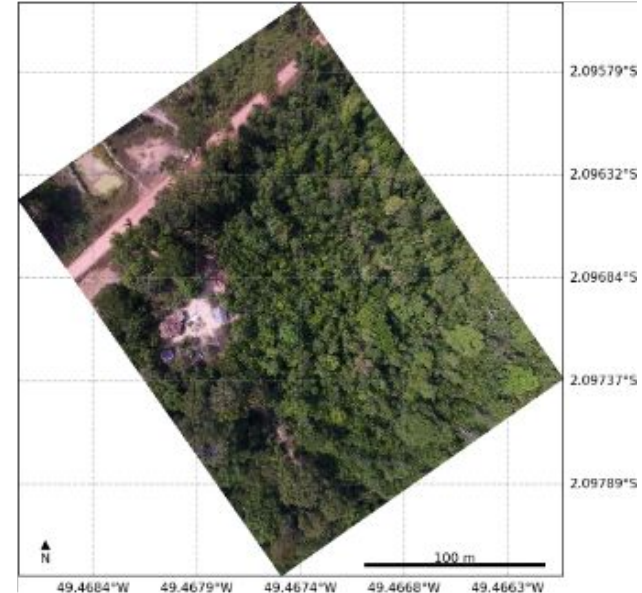
- The survey areas are mainly composed of native vegetation and water bodies, which prevents the collection of GPS control points that could be used for cartographic projection.
- The proposed solution is based on calculating the coordinates of the image corner (considered to be orientated to the North) and applying the rotation operation (affine transformation) using flight direction.



Affine Operation
Rotation

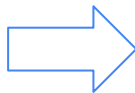


35 degrees counterclockwise
related geographic East (Course Angle)



Drone image - Example of georeferencing - City: Cametá | Date: 09/28/2019 | Flight: Ani69

Raw image over World Imagery



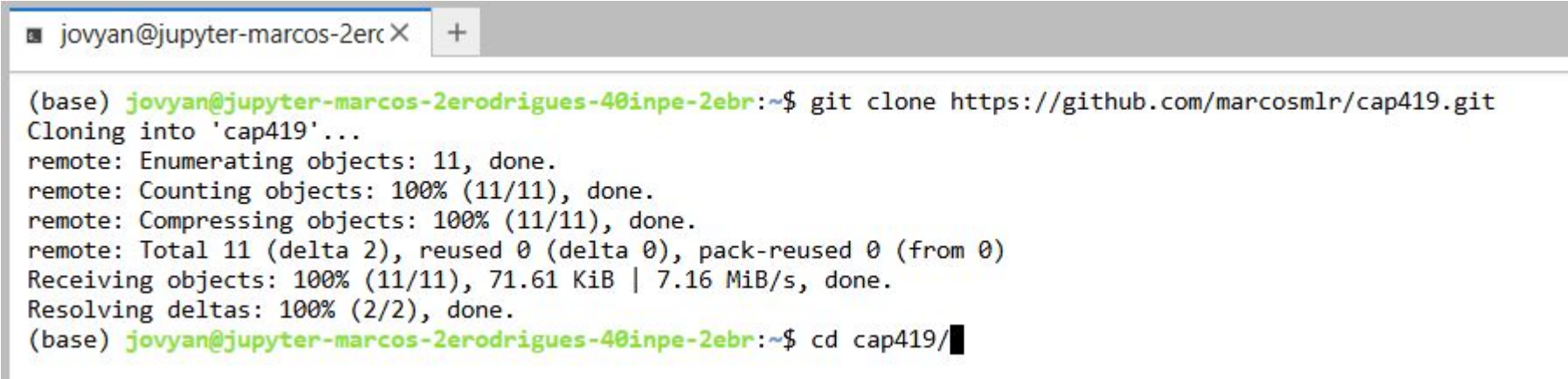
Processed Image over World Imagery



Minor positioning shifts may occur due to **the** Phantom's GPS limitations and crosswinds.

Getting Started: Clone Repository

Run in your terminal:

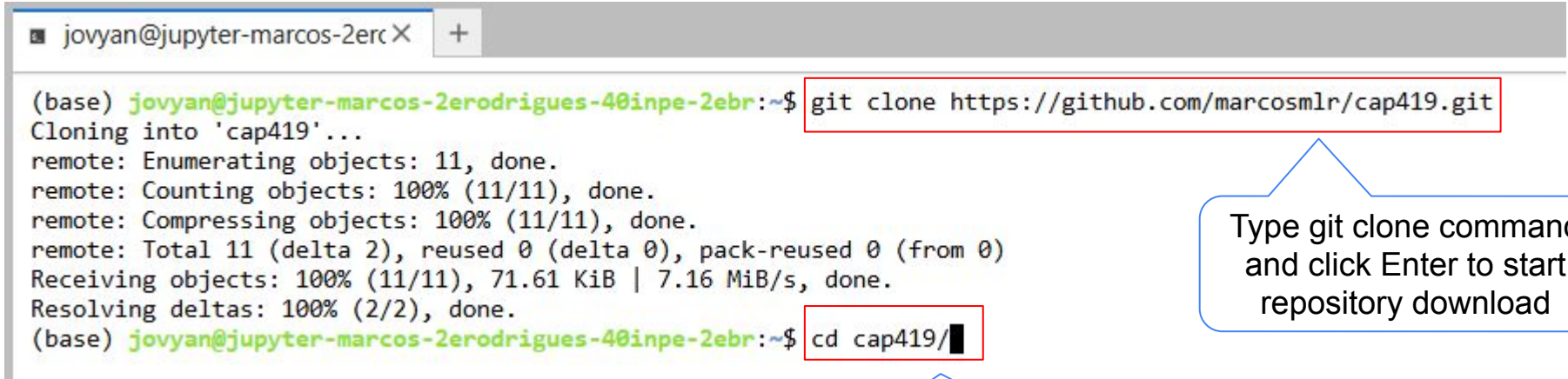


```
jovyan@jupyter-marcos-2erc X +
(base) jovyan@jupyter-marcos-2erodrigues-40inpe-2ebr:~$ git clone https://github.com/marcosmlr/cap419.git
Cloning into 'cap419'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 11 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), 71.61 KiB | 7.16 MiB/s, done.
Resolving deltas: 100% (2/2), done.
(base) jovyan@jupyter-marcos-2erodrigues-40inpe-2ebr:~$ cd cap419/
```

This will place all Jupyter notebooks on your machine (BDC-Lab).

Getting Started: Clone Repository

Run in your terminal:

A terminal window titled 'jovyan@jupyter-marcos-2erc X' with a '+' button. The terminal output shows the execution of 'git clone https://github.com/marcosmlr/cap419.git' and 'cd cap419/'. The first command is highlighted with a red box, and a callout bubble points to it with the text 'Type git clone command and click Enter to start repository download'. The second command is also highlighted with a red box, and a callout bubble points to it with the text 'After clone enter in cap419 directory to enable cube4health instalation'.

```
(base) jovyan@jupyter-marcos-2erodrigues-40inpe-2ebr:~$ git clone https://github.com/marcosmlr/cap419.git
Cloning into 'cap419'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 11 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), 71.61 KiB | 7.16 MiB/s, done.
Resolving deltas: 100% (2/2), done.
(base) jovyan@jupyter-marcos-2erodrigues-40inpe-2ebr:~$ cd cap419/
```

Type git clone command and click Enter to start repository download

After clone enter in cap419 directory to enable cube4health instalation

This will place all Jupyter notebooks on your machine (BDC-Lab).

Getting Started: Install cube4health with a single command

jovyan@jupyter-marcos-2erc X

+

```
(base) jovyan@jupyter-marcos-2erodrigues-40inpe-2ebr:~/cap419$ ./setup_cube4health.sh
```

```
=== Cloning repository ===
```

```
Cloning into 'cube4health'...
```

```
remote: Enumerating objects: 1083, done.
```

```
remote: Counting objects: 100% (312/312), done.
```

```
remote: Compressing objects: 100% (201/201), done.
```

```
remote: Total 1083 (delta 142), reused 170 (delta 81), pack-reused 771 (from 1)
```

```
Receiving objects: 100% (1083/1083), 22.18 MiB | 13.58 MiB/s, done.
```

```
Resolving deltas: 100% (603/603), done.
```

```
=== Creating Conda environment (py310) ===
```

```
Channels:
```

```
- conda-forge
```

```
Platform: linux-64
```

```
Collecting package metadata (repodata.json): done
```

```
Solving environment: done
```

```
==> WARNING: A newer version of conda exists. <==
```

```
current version: 24.11.3
```

```
latest version: 26.3.2
```

cube4health introduction

File Edit View Run Kernel Git Tabs Settings Help BDC

Filter files by name

/ cap419 /

Name	Modified
cube4health	yesterday
FieldWorkCampaigns	5 months ago
cube4health_drone_data_processing...	1 minute ago
cube4health_introduction.ipynb	15 hours ago
drone_data_harmonize_annual_me...	5/3/26, 4:08 AM - hours ago
LICENSE	yesterday
README.md	yesterday
setup_cube4health.sh	16 hours ago

cube4health_introduction.ipynb

light | Mem: 336 / 49152 MB | Disk: 1.48 / 1.97 T

Open in... Python (Geospatial)

INPE

Cube4Health

Introduction and Install

render nbviewer

Marcos L. Rodrigues

Earth Observation and Geoinformatics Division, National Institute for Space Research (INPE)
Avenida dos Astronautas, 1758, Jardim da Granja, São José dos Campos, SP 12227-010, Brazil

Contact: marcos.rodrigues@inpe.br

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Obrigado

Marcos L. Rodrigues
marcos.rodrigues@inpe.br



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